CLINICAL TRIAL FAILURES · DISCONTINUATION RATE · ONCOLOGY PHASE 2,3 · STARTS 2015–2024

PD-(L)1: 15 molecules stopped early

103 stopped trials across 15 molecules: Pembrolizumab, Nivolumab, Atezolizumab, Avelumab and 11 more. 88 of the stops were the trial's own verdict and 9 followed a decision taken elsewhere; 6 cannot be established. Against 5.2% for all oncology trials with a resolved drug target and 4.5% across all oncology Phase 2,3 trials in the same window. The rate counts trials that stopped early for efficacy, safety or benefit–risk reasons. Trials that ran to completion and missed their primary endpoint are counted separately below, never in the rate.

5.0%
103 of 2075 closed trials (95% CI 4.1%–6.0%)

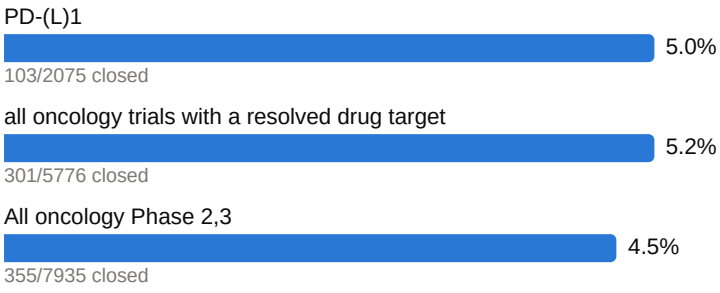
77 / 19 / 1
efficacy / safety / both (adds to 103)

36.7%
of 5655 trials have closed · 3580 still open or unresolved

1.6%
stopped for a biological reason within 3 years of starting (95% CI 1.2%–2.0%; 2669 still at risk; 1.5% area-wide)

60 of 125 **Completed and missed the primary endpoint.** Of 1469 completed trials in this segment, 125 posted a primary result on ClinicalTrials.gov that can be read — the sponsor's own comparison held to the sponsor's own threshold, or its statement that the endpoint was missed: 60 missed and 53 met, 12 split across co-primary endpoints. Counted separately and never in the rate above; a floor, not a rate. 485 of the 1351 completed trials that finished more than a year ago have posted no results at all — a gap in what can be known, not a sign of failure.

How this compares



What is in this class: 1085 distinct experimental drugs across 5655 trials (antibody 5358, small molecule 2593, unknown 965, protein 317). Most tested: PEMBROLIZUMAB (1314), NIVOLUMAB (908), DURVALUMAB (460), ATEZOLIZUMAB (425), PACLITAXEL (411). A class groups drugs by what they act on; check that the grouping is one you would make before reading the rate as a property of the mechanism.

How much does this rest on one decision? The 103 stops came from 91 sponsor-asset programmes across 67 sponsors; the largest (Roche) contributed 4; removing that programme's trials from both sides leaves 4.8%. Screened alongside 85 other segments, the rate is within what screening that many classes could produce (q=1); the molecules and trials that failed here are the record either way. A further 120 closed trials here were terminated with no cause recorded in the registry; if every one of them were biological the rate would be 10.7%. Read it as a screen worth checking against the underlying trials.

Bars show the share of closed trials stopped for biological reasons. Confidence intervals overlap where sample sizes are small — read the counts, not just the bars.

By start cohort

Trials started	Stops	Closed	Rate	95% CI
2015–2017	29	732	4.0%	2.8%–5.6%
2018–2020	49	865	5.7%	4.3%–7.4%
2021–2024	25	478	5.2%	3.6%–7.6%

Where the stops sit

Efficacy / futility: 77, Safety: 19, Benefit–risk: 6, Efficacy and safety: 1.
Sponsors with most stops: Pfizer (7), Merck & Co. (5), Roche (5), National Cancer Institute (NCI) (4), Novartis (3), Merck KGaA (3).

The stopped trials

Trial	Ph	Sponsor	Experimental drugs	Stopped	Type	Registry stop reason
NCT05860296	2	SignalChem Lifesciences Corporation	SLC-391, PEMBROLIZUMAB	2025-12	Efficacy	Lack of efficacy and enrollment challenges

Trial	Ph	Sponsor	Experimental drugs	Stopped	Type	Registry stop reason
NCT03274258	2	M.D. Anderson Cancer Center	IPILIMUMAB, NIVOLUMAB	2025-11	Efficacy	The study met its prespecified futility criteria during the prespecified interim futility analysis after enrolling the first cohort of 10 patients.
NCT03225547	2	University of Chicago	PEMBROLIZUMAB, MIFEPRISTONE	2025-09	Safety	side effects (rash)
NCT05491317	2	Genmab GMAB	Tecaginlimab, PEMBROLIZUMAB	2025-08	Efficacy	After reviewing all available data from Part 1 of the trial, the Sponsor has decided not to proceed to Part 2 (randomized Phase 2) of the trial due to lack of efficacy
NCT04955743	2	Yale University	PEMBROLIZUMAB, LENVATINIB	2025-05	Efficacy	Lack of activity in the drugs
NCT05469178	2	BerGenBio ASA	BEMCENTINIB, PEMBROLIZUMAB, PEMETREXED, CARBOPLATIN	2025-04	Efficacy	Lack of efficacy

The 6 most recent of 103 stopped trials. Every molecule behind all 103 is named above; the complete trial list, the trials still running and the terminations with no readable cause ship with the dataset and the diligence report.

Method

Denominator: ClinicalTrials.gov interventional Phase 2,3 oncology trials started 2015–2024 that have closed (completed or terminated). Numerator: terminated trials whose registry stop reason is classified as biological (efficacy, safety or benefit–risk) by Classification V2 — held-out precision 95.5%, recall 95.3% (n=600). Drugs are linked to ChEMBL and the NCI Thesaurus; 76% of industry oncology trials in this window carry a resolved drug target, and a trial without one cannot enter a mechanism class. Intervals are Wilson 95%.

Limits

Not a failure rate: completed trials that missed their endpoints are not in the rate. They are counted separately where the sponsor posted a primary result that can be read — most did not, so that count is a floor — and programmes discontinued after a completed trial do not appear. Stop reasons are sponsor-reported. This is a closed-trial proportion, not a time-to-event analysis: only 36.7% of trials in this segment have closed, and a trial that stops early enters the denominator sooner than one that runs to completion, which can inflate the rate in immature segments. Recent cohorts have fewer closed trials, so their rates are less stable. Research signals, not clinical or investment advice.

The whole cohort, trial by trial. The diligence report names every trial behind these figures — the ones still running and the ones that completed and missed their primary endpoint included — and places your molecule against each one that failed. Free during the beta, in exchange for a work email, delivered immediately: **clinicaltrialfailures.com/asset-check**

Sources: ClinicalTrials.gov (NLM); ChEMBL (EMBL-EBI, CC BY-SA 3.0); NCI Thesaurus (NCI); RxNorm/RxClass (NLM); SEC EDGAR. Classification, linkage and rates by Clinical Trial Failures. Rebuilt weekly; this brief covers trials started 2015–2024.